Mutual Fund Analytics

Data Analyst Internship Project

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Finally, I would like to acknowledge everyone who contributed directly or indirectly to the successful completion of this project.

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Executive Summary

The Mutual Fund Analytics Platform was developed as part of the Data Analyst Internship at Bluestock Fintech with the objective of transforming fragmented mutual fund data into meaningful business insights through data engineering, financial analytics, and interactive business intelligence dashboards. The project aimed to provide investors and financial analysts with a centralized platform for evaluating mutual fund performance, tracking industry trends, and understanding investor behavior.

The project involved the collection and integration of multiple datasets, including Fund Master Data, NAV History, Benchmark Indices, Category Inflows, Investor Transactions, and Industry Folio Count data. A robust ETL (Extract, Transform, Load) pipeline was developed using Python, Pandas, and SQL to clean, transform, and prepare the data for analysis. The processed datasets were stored in a structured database and further analyzed using statistical and financial techniques.

Exploratory Data Analysis (EDA) was conducted to identify industry trends and investment patterns. The analysis revealed that the mutual fund ecosystem covered in this project manages approximately ₹62.74 Lakh Crore in Assets Under Management (AUM), with ₹15.90 Lakh Crore in SIP inflows, 26.12 Crore investor folios, and 40 mutual fund schemes. Category-wise analysis showed that Liquid Funds recorded the highest net inflow of ₹4,51,275 Crore during FY25, indicating strong investor preference for liquidity and capital preservation. Advanced performance analytics were performed using historical NAV data to calculate key financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Tracking Error.

Among the evaluated schemes, ICICI Prudential Liquid Fund – Regular Growth achieved the highest Sharpe Ratio of 7.68, while SBI Small Cap Fund – Regular Plan Growth recorded the highest 3-Year Return of 23.39%. Additionally, Mirae Asset Emerging Bluechip Fund – Regular Growth maintained the highest Assets Under Management of ₹49,046 Crore.

The final outcome of the project was a comprehensive Power BI dashboard consisting of Industry Overview, Market Trends, Fund Performance, and

Investor Analytics modules. The platform successfully transformed raw financial data into actionable insights, enabling data-driven investment analysis and supporting informed decision-making for investors and financial institutions.

Company Overview

Introduction to Bluestock Fintech

Bluestock Fintech is a financial technology company that focuses on simplifying investment analytics and financial decision-making through data-driven solutions. The organization leverages modern technologies such as Data Analytics, Business Intelligence, Financial Modeling, and Automation to provide actionable insights for investors, financial advisors, and institutions.

The company aims to bridge the gap between complex financial information and practical investment decisions by developing analytical platforms that enable users to evaluate market trends, monitor financial performance, and identify investment opportunities.

Business Domain

Bluestock Fintech operates in the Financial Technology (FinTech) sector, which combines finance and technology to deliver innovative financial services. The company focuses on investment analytics, financial data management, and business intelligence solutions that help users make informed decisions using real-time and historical financial data.

Services and Solutions

The company provides solutions in the following areas:

• Financial Data Analytics

• Investment Research

• Mutual Fund Analysis

- Business Intelligence Dashboards
- Market Trend Analysis
- Portfolio Monitoring
- Data Visualization and Reporting

Role of Data Analytics in Bluestock Fintech

Data analytics plays a critical role in transforming raw financial data into meaningful business insights. Through data engineering, statistical analysis, and visualization techniques, large volumes of financial data can be processed and interpreted efficiently.

For this internship project, data analytics was used to:

- Integrate mutual fund datasets from multiple sources.
- Analyze Assets Under Management (AUM) trends.
- Evaluate mutual fund performance using risk-adjusted metrics.
- Study investor transaction behavior.
- Generate interactive dashboards for business users.

Internship Project Context

As part of the Data Analyst Internship Program at Bluestock Fintech, a Mutual Fund Analytics Platform was developed to address challenges associated with fragmented financial data and limited access to consolidated investment insights.

The project integrated multiple mutual fund datasets, performed advanced financial analysis, and delivered interactive Power BI dashboards capable of supporting data-driven investment decisions. The platform enables users to monitor industry trends, compare fund performance, evaluate risk-return characteristics, and understand investor behavior through comprehensive analytics and visualization.

Project Objective

Project Background

The Indian mutual fund industry has experienced significant growth over the past decade, resulting in an increasing volume of financial data generated across various platforms. However, mutual fund information such as NAV history, Assets Under Management (AUM), benchmark indices, category inflows, and investor transaction records are often available in separate datasets, making comprehensive analysis difficult for investors and financial institutions.

The absence of a centralized analytics platform creates challenges in evaluating fund performance, comparing investment options, tracking industry trends, and understanding investor behavior. As a result, investors often rely on fragmented information and manual analysis, which may lead to inefficient decision-making.

Problem Statement

Investors and financial analysts require a unified platform that can consolidate mutual fund data, perform advanced financial analysis, and provide meaningful insights through interactive dashboards.

The key challenges addressed in this project include:

• Data fragmentation across multiple financial datasets.

• Difficulty in comparing mutual fund performance using risk-adjusted metrics.

• Lack of benchmark-based performance evaluation.

• Limited visibility into investor behavior and transaction patterns.

• Dependence on static reports instead of interactive analytics dashboards.

Project Objectives

The primary objective of this project was to design and develop a comprehensive Mutual Fund Analytics Platform capable of transforming raw financial data into actionable business insights.

The specific objectives of the project were:

Objective 1: Build an ETL Pipeline

Develop an automated ETL (Extract, Transform, Load) process to collect, clean, transform, and integrate mutual fund datasets from multiple sources.

Objective 2: Design a Structured Database

Create a relational database schema capable of storing mutual fund information, benchmark data, investor transactions, and performance metrics in a structured format.

Objective 3: Perform Exploratory Data Analysis

Analyze historical mutual fund data to identify trends, patterns, anomalies, and relationships across different financial indicators.

Objective 4: Calculate Performance and Risk Metrics

Compute key financial metrics including:

• CAGR (Compound Annual Growth Rate)

• Sharpe Ratio

• Sortino Ratio

• Alpha

• Beta

• Maximum Drawdown

• Tracking Error

These metrics enable comprehensive evaluation of mutual fund performance from both return and risk perspectives.

Objective 5: Develop Interactive Dashboards

Build dynamic Power BI dashboards that provide visual insights into industry growth, fund performance, market trends, and investor behavior.

Objective 6: Analyze Investor Behavior

Study investor transaction patterns across different age groups, transaction types, and geographic regions to identify behavioral trends.

Objective 7: Benchmark Fund Performance

Compare mutual fund performance against benchmark indices such as NIFTY50 and NIFTY100 to evaluate relative performance and market exposure.

Objective 8: Generate Actionable Business Insights

Transform analytical findings into meaningful recommendations that can support investment decision-making and strategic planning.

Project Scope

The scope of the project includes data integration, financial analytics, dashboard development, and investor behavior analysis within the mutual fund domain.

The project covers:

• 40 Mutual Fund Schemes

• Historical NAV Analysis

• Benchmark Comparison

• Category-wise Inflow Analysis

• Investor Transaction Analysis

• Risk-Return Evaluation

• Business Intelligence Reporting

The platform is designed to provide investors, analysts, and financial institutions with a centralized environment for mutual fund analysis and decision support.

Expected Outcomes

The successful implementation of this project is expected to provide:

• Improved access to mutual fund analytics.

• Faster and more accurate investment analysis.

• Enhanced understanding of industry trends.

• Better evaluation of risk-adjusted returns.

• Data-driven investment decision support.

• Interactive visualization of financial performance and investor behavior.

The final outcome is a comprehensive Mutual Fund Analytics Platform that integrates data engineering, financial analytics, and business intelligence into a single analytical solution.

Methodology & Tech Stack

Methodology

The Mutual Fund Analytics Platform was developed using a structured data analytics workflow consisting of data collection, ETL processing, financial analysis, and dashboard visualization. The objective was to transform raw mutual fund datasets into meaningful business insights for performance evaluation and investor analysis.

The project workflow followed the sequence:

Data Collection → ETL Processing → Database Storage → Exploratory Data Analysis → Performance Analytics → Dashboard Development

Data Sources

The analysis was performed using multiple financial datasets, including Fund Master, NAV History, Benchmark Indices, Category Inflows, Investor Transactions, and Industry Folio Count data.

The Fund Master dataset served as the primary reference table, while NAV History and Benchmark datasets were used for return and risk calculations. Investor transaction and category inflow datasets supported behavioral and market trend analysis.

ETL Process

Python and Pandas were used to build the ETL pipeline.

Extract

Raw CSV files were imported and validated to ensure consistency across datasets.

Transform

Data preprocessing activities included:

Handling missing values

Removing duplicate records

Standardizing date formats

Creating derived analytical features

Daily returns were calculated from NAV values and benchmark returns were integrated for comparative performance analysis.

Load

The processed datasets were stored in structured tables and exported for further analysis and dashboard development.

Financial Analytics

Several performance and risk metrics were calculated using historical NAV data.

CAGR (Compound Annual Growth Rate)

Measures annualized growth over a specified investment period.

Sharpe Ratio

Evaluates risk-adjusted returns by comparing excess returns against total volatility.

Sortino Ratio

Measures returns relative to downside risk.

Alpha and Beta

Alpha measures benchmark outperformance, while Beta measures market sensitivity.

Maximum Drawdown

Identifies the largest decline from a historical peak.

Tracking Error

Measures deviation between fund returns and benchmark returns.

These metrics provided a comprehensive assessment of mutual fund performance.

Dashboard Development

Power BI was used to develop four interactive dashboards:

Industry Overview

SIP & Market Trends

Fund Performance Analytics

Investor Analytics

The dashboards enable users to analyze industry growth, compare fund performance, evaluate risk-return characteristics, and understand investor behavior through interactive visualizations.

Technology Stack

Technology	**Purpose**
Python	Data Processing & Analytics
Pandas	ETL & Data Transformation
NumPy	Numerical Computation
SciPy	Statistical Analysis
PostgreSQL	Data Storage
SQL	Querying & Validation

Power BI	Dashboard Development
Git & GitHub	Version Control

The combination of these technologies enabled the development of a scalable analytics platform capable of transforming financial data into actionable business insights.

Implementation & Analysis

Data Preparation and Processing

The first stage of implementation involved collecting and integrating multiple mutual fund datasets into a unified analytical environment. Raw datasets contained historical NAV records, benchmark index values, category inflows, investor transactions, and fund master information.

Data preprocessing was performed using Python and Pandas. Missing values were identified and handled appropriately, duplicate records were removed, and data types were standardized to ensure consistency across datasets. Date fields were converted into a uniform datetime format to support time-series analysis.

To support financial analytics, benchmark index data was merged with NAV history data using the transaction date as the common key. This integration enabled the calculation of benchmark-relative performance metrics such as Alpha, Beta, and Tracking Error.

Return Calculation

Daily returns were calculated from historical NAV values using percentage change methodology.

The daily return calculation enabled the transformation of raw NAV data into a format suitable for risk-return analysis and performance evaluation.

The generated return series served as the foundation for all subsequent financial metrics.

CAGR Analysis

Compound Annual Growth Rate (CAGR) was calculated to measure the annualized growth of mutual fund investments over different periods.

The analysis provided a standardized measure of long-term fund performance and enabled comparison across schemes with different return patterns.

Results indicated that SBI Small Cap Fund – Regular Plan Growth achieved the highest 3-Year Return of 23.39%, demonstrating strong long-term growth potential among the evaluated schemes.

Risk-Adjusted Performance Analysis

Traditional return measures do not account for the level of risk taken by investors. Therefore, risk-adjusted metrics were calculated to evaluate the efficiency of fund performance.

Sharpe Ratio

The Sharpe Ratio measures excess return generated per unit of total risk.

Analysis revealed that ICICI Prudential Liquid Fund – Regular Growth recorded the highest Sharpe Ratio of 7.68, indicating superior risk-adjusted performance compared to other schemes in the dataset.

A higher Sharpe Ratio suggests that the fund generated relatively higher returns while maintaining lower volatility.

Sortino Ratio

The Sortino Ratio was calculated to evaluate returns relative to downside risk. Unlike the Sharpe Ratio, it focuses only on negative return volatility and provides a more targeted measure of investment risk.

The metric helped identify funds capable of generating stable returns while minimizing downside exposure.

Benchmark Performance Analysis

Benchmark indices were incorporated into the analysis to evaluate mutual fund performance relative to market movements.

NIFTY50 and NIFTY100 indices were used as benchmark references for comparison.

Alpha

Alpha was calculated to measure excess return generated by a fund beyond benchmark expectations.

Positive Alpha values indicate benchmark outperformance, while negative values indicate underperformance.

Beta

Beta was calculated to measure the sensitivity of mutual fund returns to market movements.

Funds with higher Beta values tend to exhibit greater volatility relative to benchmark indices, whereas lower Beta values indicate reduced market exposure.

Drawdown and Tracking Error Analysis

Maximum Drawdown

Maximum Drawdown was calculated to identify the largest decline experienced by a fund from its historical peak value.

This metric provided insights into downside risk and helped evaluate capital preservation capability.

Tracking Error

Tracking Error measured the deviation between mutual fund returns and benchmark returns.

The metric was used to assess the consistency of benchmark tracking and identify funds with significant performance divergence.

Fund Ranking and Scorecard Development

To facilitate objective comparison across schemes, a composite fund scoring framework was developed.

The scoring model incorporated:

CAGR

Sharpe Ratio

Alpha

Expense Ratio

Maximum Drawdown

Each metric was ranked and assigned weighted importance. The weighted scores were combined to generate a normalized fund score ranging from 0 to 100.

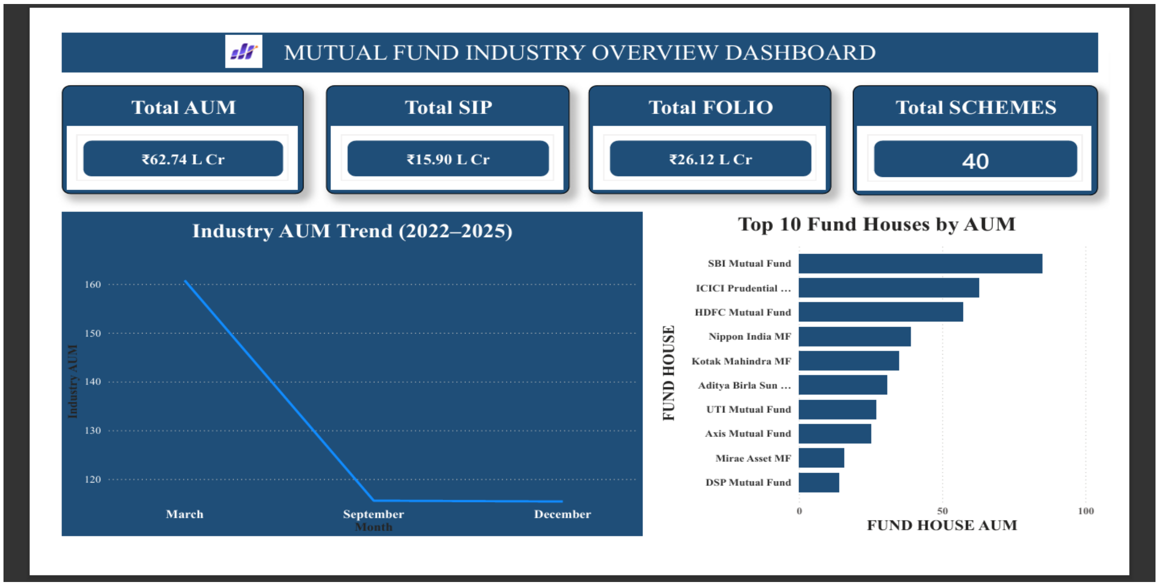
This approach enabled comprehensive comparison of mutual funds based on both return generation and risk management characteristics.

Dashboard Analysis

The processed datasets and analytical outputs were integrated into Power BI dashboards for visualization and business intelligence reporting.

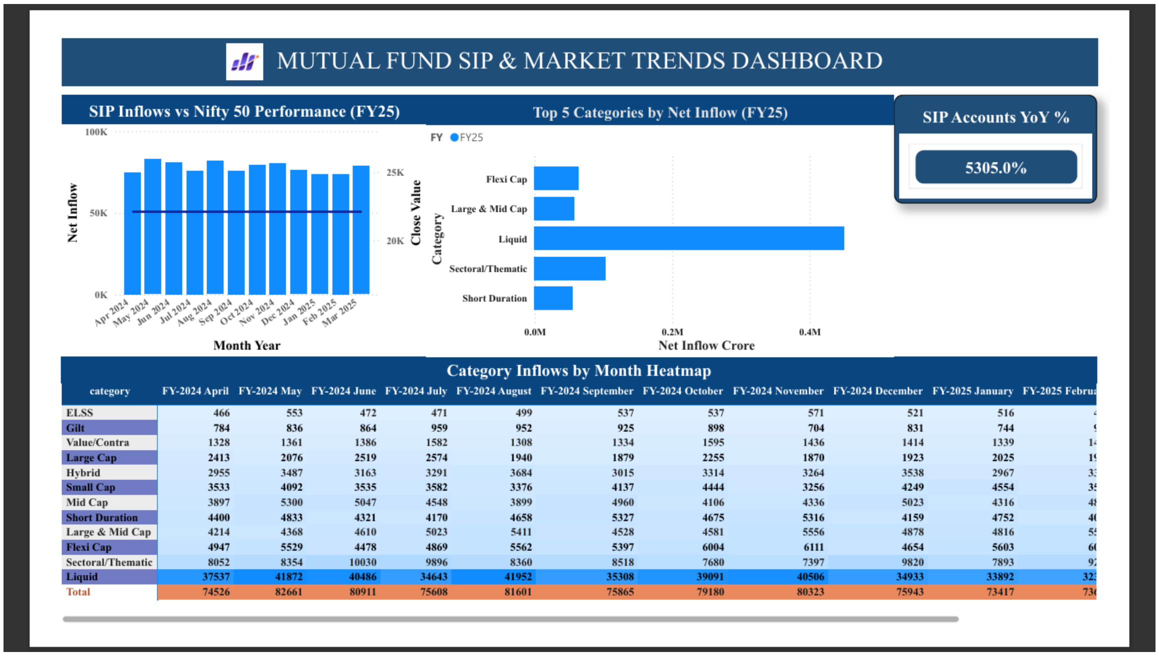
Industry Overview Dashboard

These indicators highlight the scale of the mutual fund ecosystem analyzed within the project.



SIP & Market Trends Dashboard

Category-wise inflow analysis showed that Liquid Funds recorded the highest net inflows of ₹4,51,275 Crore during FY25, reflecting strong investor preference for liquidity and low-risk investment options.



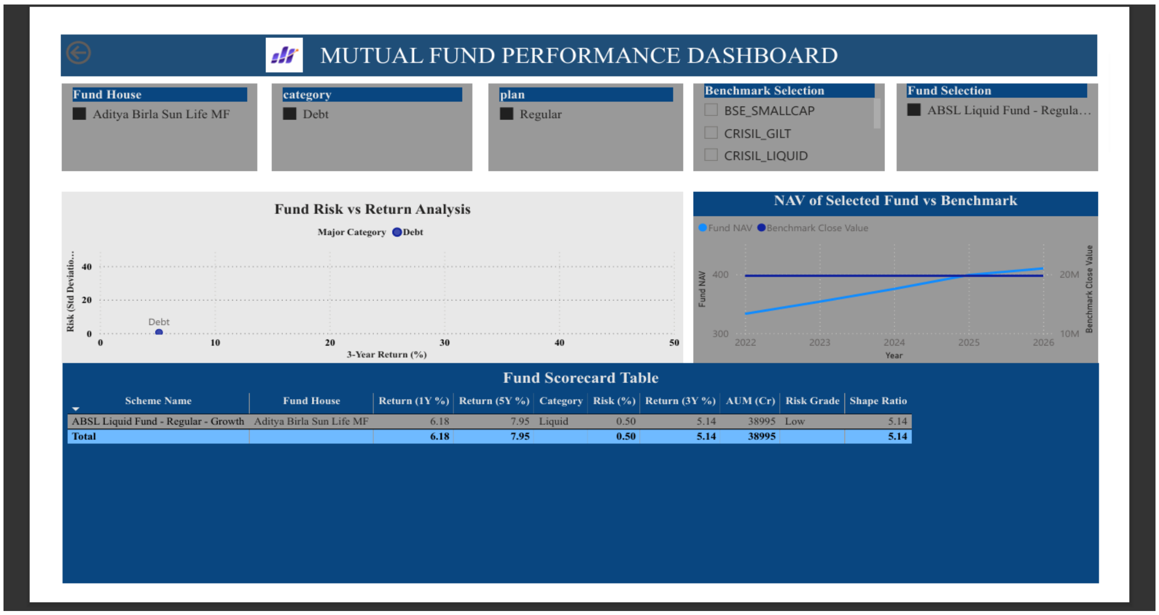
Fund Performance Dashboard

The performance dashboard revealed that:

ICICI Prudential Liquid Fund – Regular Growth achieved the highest Sharpe Ratio of 7.68.

SBI Small Cap Fund – Regular Plan Growth recorded the highest 3-Year Return of 23.39%.

Mirae Asset Emerging Bluechip Fund – Regular Growth maintained the highest AUM of ₹49,046 Crore.



Investor Analytics Dashboard

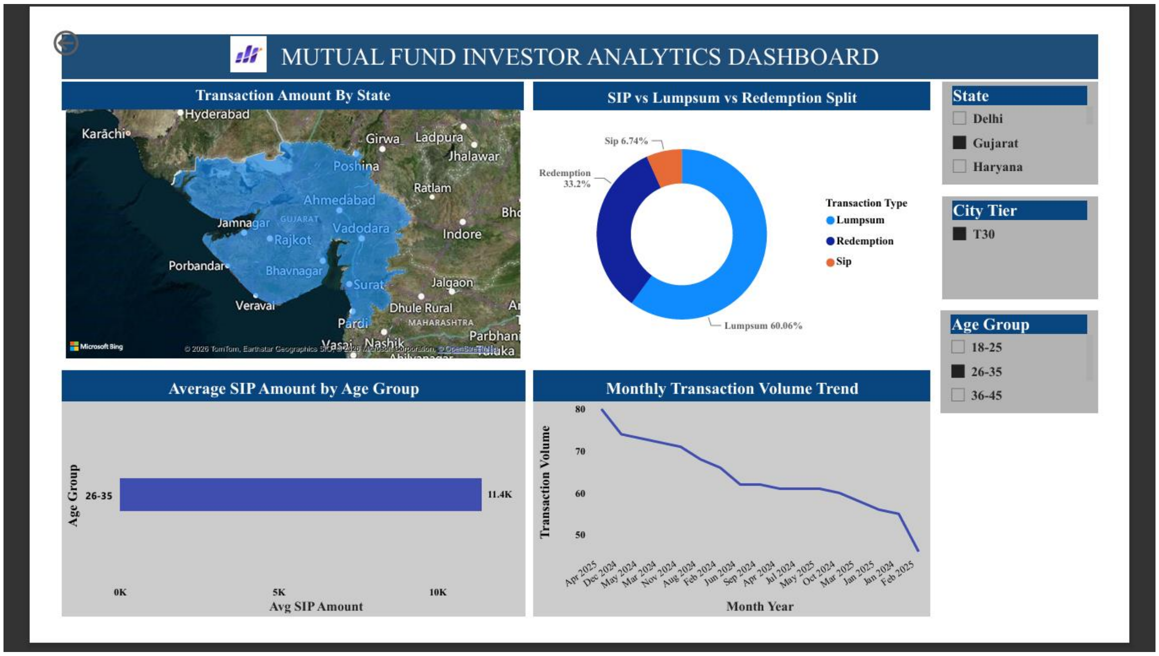
Investor behavior analysis showed that:

Lumpsum investments accounted for 58.49% of transactions.

Investors aged 56 years and above recorded the highest participation with 13.6K transactions.

Punjab generated the highest transaction amount of approximately ₹315.78 million.

These findings provide valuable insights into investor preferences, demographic trends, and regional participation patterns.



Key Findings & Results

The analysis of mutual fund industry data, investor transactions, benchmark indices, and scheme performance generated several important business insights.

Industry Growth Findings

The Industry Overview Dashboard indicated strong growth and participation within the mutual fund ecosystem.

Key observations include:

Total Assets Under Management (AUM) reached ₹62.74 Lakh Crore.

SIP inflows amounted to ₹15.90 Lakh Crore.

The industry maintained 26.12 Crore investor folios.

A total of 40 mutual fund schemes were analyzed.

These figures demonstrate the increasing adoption of mutual funds as a preferred investment vehicle among Indian investors.

Category-Wise Investment Trends

Analysis of category inflows revealed significant differences in investor preferences.

Key findings:

Liquid Funds recorded the highest net inflow of ₹4,51,275 Crore during FY25.

Investors showed a strong preference for liquid and low-risk investment options.

Category inflow patterns indicated that investment decisions are influenced by liquidity requirements and risk appetite.

The dominance of Liquid Funds suggests that investors prioritize capital preservation and flexibility during uncertain market conditions.

Fund Performance Findings

Risk-return analysis highlighted the importance of evaluating mutual funds using multiple performance indicators.

Major findings include:

ICICI Prudential Liquid Fund – Regular Growth achieved the highest Sharpe Ratio of 7.68.

SBI Small Cap Fund – Regular Plan Growth generated the highest 3-Year Return of 23.39%.

Mirae Asset Emerging Bluechip Fund – Regular Growth maintained the highest Assets Under Management (AUM) of ₹49,046 Crore.

These results indicate that different funds excel across different dimensions such as risk-adjusted returns, growth performance, and investor trust.

Investor Behavior Findings

Investor transaction analysis provided valuable insights into demographic and geographic participation.

Key observations include:

Lumpsum investments accounted for 58.49% of total transactions.

Investors aged 56 years and above recorded the highest participation with 13.6K transactions.

Punjab generated the highest transaction amount of approximately ₹315.78 Million.

These findings suggest that experienced investors continue to play a major role in mutual fund investments and that regional participation varies significantly across India.

Conclusion & Future Scope

Conclusion

The Mutual Fund Analytics Platform was successfully developed to analyze mutual fund performance, industry trends, and investor behavior. Multiple datasets including NAV history, benchmark indices, category inflows, and investor transactions were integrated using an ETL pipeline built with Python.

Various financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Tracking Error were calculated to evaluate fund performance. Interactive Power BI dashboards were created to visualize industry growth, fund performance, market trends, and investor insights.

The analysis identified key findings such as Total AUM of ₹62.74 Lakh Crore, SIP Inflows of ₹15.90 Lakh Crore, highest category inflows in Liquid Funds, and significant investor participation from the 56+ age group. The project successfully transformed raw financial data into actionable business insights and supported data-driven investment analysis.

Recommendations

Based on the analysis performed in this project, the following recommendations are proposed:

1. Investors should evaluate mutual funds using risk-adjusted metrics such as Sharpe Ratio and Sortino Ratio rather than relying solely on returns.
2. Fund houses should closely monitor category-wise inflow trends to identify changing investor preferences and optimize product offerings.
3. Benchmark-based performance evaluation should be adopted to measure fund effectiveness relative to market indices.
4. Investor awareness programs can be targeted towards younger age groups to improve mutual fund participation across demographics.
5. The dashboard should be integrated with automated data refresh mechanisms to provide real-time investment insights.

6. Fund managers should regularly track Alpha, Beta, and Tracking Error metrics to improve portfolio performance and risk management.
7. Business intelligence dashboards should be used as a decision-support tool for investment analysis and strategic planning.

Limitations

Despite providing valuable insights, the project has certain limitations:

1. The analysis is based on historical mutual fund data and does not guarantee future performance.
2. Only selected mutual fund schemes and benchmark indices were included in the study.
3. Real-time market data integration was not implemented within the project scope.
4. Investor demographic information available in the dataset was limited.
5. External factors such as economic conditions, interest rates, inflation, and geopolitical events were not incorporated into the analysis.
6. The dashboard relies on periodic dataset updates and may not reflect live market movements.
7. Machine learning and predictive forecasting models were not included in the current implementation.

References

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